

Green Cycle - Scrum Basics

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1. Sprint 0: Planning & Setup

1.1. Project Overview

Green Cycle is a web platform for sharing and discovering upcycling projects. The development is split into four phases aligned with the milestones defined in the *Green Cycle - Project Charter* document, specifically *backend architecture, primary features, secondary features, and final testing and deployment*. This document covers the planning decisions made in Sprint 0, the tooling we will use throughout the project, and the task breakdown for Sprint 1.

1.2. Team & Roles

Member	Scrum Role	Primary Responsibility
Boris Marinković	Product Owner / Developer	Requirements, backend architecture, database schema
Janko Kondić	Scrum Master / Developer	Sprint coordination, backend features, deployment
Nikola Gostovikj	Developer	Frontend (React), UI components, testing

1.3. Tools

After reviewing the options (*Trello, Jira, OpenProject, Obsidian with Kanban plugin*), we decided to use the following:

- **GitHub Projects:** For the Scrum board and sprint management. Since we already have a GitHub repository, using GitHub Projects keeps everything in one place and avoids switching tools.
- **GitHub Issues:** For individual task tracking.
- **GitHub Milestones:** For sprint boundaries. Each sprint corresponds to a milestone with a due date.
- **Discord:** For daily async communication.
- **Figma:** For prototype and UI design work (already in use from previous phases).

1.4. Sprint Structure

Sprints are 2 weeks long. Given the 4-month project window and the milestones defined in the charter, we are planning the following sprint structure:

Sprint	Focus	Target Deadline
<i>Sprint 1</i>	Database schema, user roles, authentication	End of March 2026
<i>Sprint 2</i>	Project creation, materials management, search and filtering	End of April 2026
<i>Sprint 3</i>	Comments, private messaging, achievements, content reporting	End of May 2026
<i>Sprint 4</i>	Bug fixes, testing, deployment, documentation	End of June 2026

1.5. Complexity Estimation

We are using the Fibonacci scale for story point estimation: 1, 2, 3, 5, 8, 13. A story point of 1 means a very simple change that takes under an hour, 13 means a large, complex feature that probably needs to be broken down further. Estimates were agreed on by the team using a quick discussion for each task.

1.6. Definition of Done

A task is considered done when the code is pushed to the repository, reviewed by at least one other team member, and the feature works correctly in the development environment without breaking existing functionality.

2. Sprint 1: Backend Architecture & Authentication

Sprint 1 covers the first project milestone: database schema, backend structure, and user role management. The goal at the end of this sprint is to have a working backend with user registration, login, and role-based access control in place.

Sprint dates: March 23 - April 6, 2026

2.1. Sprint 1 Backlog

#	Task	Story Points	Assignee	User Story
1	Set up project repository structure	1	Janko	Foundation
2	Design and implement the PostgreSQL database schema	8	Boris	All user stories depend on this
3	Set up Node.js backend with Express and PostgreSQL connection	3	Janko	Foundation
4	Implement user registration endpoint with input validation	3	Boris	TC6.1, TC7.4
5	Implement login endpoint with JWT authentication	5	Boris	TC7.4
6	Implement role-based access control middleware (user, moderator, admin)	5	Janko	Needed for US7, US8, admin panel
7	Set up React project with routing and basic layout skeleton	3	Nikola	Foundation for all frontend network
8	Create registration and login pages (frontend)	3	Nikola	TC6.1
9	Connect frontend auth forms to backend endpoints	2	Nikola	TC6.1
10	Write basic API documentation for auth endpoints	1	Boris	Required by charter (section 6.5)

3. Sprints 2-4: Overview

The following is a high-level overview of planned work for the remaining sprints. Detailed task breakdowns will be prepared at the start of each sprint.

3.1. Sprint 2: Primary Features (~38 story points)

- Project submission form (TC2.4, TC3.1, TC6.2).
- Material management and tagging system (TC9.3, TC9.4).
- Search and filtering by material, difficulty, age group, and cost (TC1.1-1.4, TC4.1-4.3).
- Project detail page (TC2.1-2.3).
- Basic user profile page (TC7.1-7.3).

3.2. Sprint 3: Secondary Features (~32 story points)

- Comment system (TC3.2, TC3.3).
- Private messaging (TC5.1-5.4).
- Achievement and points system (TC3.4, TC6.4).
- Content reporting and moderation tools.
- Profile editing and external link support (TC8.1).

3.3. Sprint 4: Testing, Fixes, Deployment (~20 story points)

- End-to-end testing of all main user flows.
- Bug fixes from Sprint 3 review.
- Production deployment and environment configuration.
- Final documentation (setup guide, usage guide, known bugs list).

4. Burndown Chart

We will maintain a burndown chart throughout the project using GitHub's built-in sprint burndown report available through GitHub Projects. At the start of each sprint, the total story points are logged. After each day with completed tasks, the remaining points are updated.

The burndown chart for Sprint 1 will be shared at the Sprint 1 review meeting. The ideal burndown line starts at 34 points on March 23 and reaches 0 by April 6.

Sprint	Total Points	Ideal End Date	Status
<i>Sprint 1</i>	34	April 6, 2026	<i>In progress</i>
<i>Sprint 2</i>	~38	April 30, 2026	<i>Planned</i>
<i>Sprint 3</i>	~32	May 31, 2026	<i>Planned</i>
<i>Sprint 4</i>	~20	June 30, 2026	<i>Planned</i>